Q.P Code **D 123237**	Total Pages **3**	Name **604763**
		Register No.

SECOND SEMESTER (CUFYUGP) DEGREE EXAMINATION, APRIL 2025

MATHEMATICS

MAT2MN102 Calculus and Matrix Algebra

2024 Admission Onwards

Maximum Time :2 Hours	**Maximum Marks :70**

Section A

All Question can be answered. Each Question carries 3 marks (Ceiling : 24 Marks)

1	Find $\int \frac{\sin x}{\cos^2 x}\, dx$
2	Find $\int \sin^3 x\, dx$
3	Find the average value of the function $f(x) = \frac{1}{x}$ over the interval $[1, e]$
4	Evaluate $\int_0^4 3x\sqrt{25 - x^2}\, dx$
5	Find $\int xe^x\, dx$
6	Describe the level surfaces of $f(x, y, z) = z^2 - x^2 - y^2$
7	Explain the domain of $f(x, y) = \ln xy$
8	For $n \times m$ matrices $\mathbf{A}$ and $\mathbf{B}$, show that $(\mathbf{A} + \mathbf{B})^{\mathbf{T}} = \mathbf{A}^{\mathbf{T}} + \mathbf{B}^{\mathbf{T}}$
9	True or False:"For any $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$, we get $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \mathbf{A^2} - \mathbf{B^2}$". Explain your answer
10	Find the eigenvalues and eigenvectors of the matrix $\begin{pmatrix} -8 & 1 \\ 16 & 0 \end{pmatrix}$

Section B

All Question can be answered. Each Question carries 6 marks (Ceiling : 36 Marks))

11	Evaluate the integral $\int_1^7 f(x)\,dx$, if $f(x) = \begin{cases} x^2 & 1 \le x < 3 \\ 2x+2 & 3 \le x < 5 \\ x-1 & 5 \le x \le 7 \end{cases}$
12	Using definite integral, find the area of the region below the curve $y = x - x^2$ and above the x- axis.
13	Find the area of the region enclosed by the curves $$y = x^3 - 4x, y = 0, x = 0, x = 2$$
14	Evaluate $\int e^x \cos x\,dx$
15	Determine whether the limit exists. If so, find its value. $$\lim_{(x,y)\to(0,0)} \frac{\cos xy}{x^2+y^2}$$
16	If, $f(x,y,z) = \tan^{-1}\left(\frac{1}{xy^2z^3}\right)$, then find f_x, f_y and f_z
17	Solve the following system of linear equations $x_1 + 2x_2 + 2x_3 = 2$ $x_1 + x_2 + x_3 = 0$ $x_1 - 3x_2 - x_3 = 0$
18	Construct an orthogonal matrix from the eigenvectors of the symmetric matrix $\begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}$

Section C

Answer any ONE. Each Question carries 10 marks (1x10=10 Marks))

19	Evaluate the integral $\int_0^3 \sqrt{6x - x^2}\, dx$ by completing the square and applying appropriate formulas from geometry.
20	Let $f(x, y, z) = \sqrt{x^2 - 2y^2 + 4z^2}$, find 1. $\left.\frac{\partial f}{\partial x}\right\|_{(2,1,1)}$ 2. $\left.\frac{\partial f}{\partial y}\right\|_{(1,2,3)}$ 3. $\left.\frac{\partial f}{\partial z}\right\|_{(2,1,3)}$